

CCNA 200-301

CHAPTER 30

TFTP Backup & Restore

Backing up and restoring configuration files and IOS images on Cisco routers and switches

Scope: This chapter follows the provided video content and command sequence for TFTP backup and restore.

Main topics: Running-config backup, running-config restore, IOS image backup, and verification of the IOS filename in flash.

Chapter Objectives

Understand how to copy a Cisco device's running configuration to a TFTP server, restore a saved configuration from TFTP, identify the IOS image filename stored in flash memory, and copy that IOS image to a TFTP server.

1. What the Video Covers

The lesson demonstrates using a TFTP server as an external location for backups. Two kinds of data are handled: configuration files and the IOS image.

Backup / Restore task	Purpose	Main command
Back up running configuration	Send the current running configuration to the TFTP server.	<code>copy running-config tftp:</code>
Restore configuration	Load a saved configuration from TFTP into the running configuration.	<code>copy tftp: running-config</code>
Check IOS filename	Identify the exact IOS image file stored in flash.	<code>show flash</code>
Back up IOS image	Copy the IOS image from flash to the TFTP server.	<code>copy flash: tftp:</code>

Core idea: TFTP is used as an external server location so configuration or IOS files are not kept only on the network device.

2. Backing Up the Running Configuration

The running configuration is the configuration currently active on the device. The video copies it to the external TFTP server.

```
copy running-config tftp:
```

After entering the command, the device prompts for two important pieces of information:

1. Server IP Address - the address of the TFTP server. The example in the video is 192.168.1.50.
2. Destination Filename - the name under which the backup will be stored on the TFTP server, such as august_2020.

Command flow

```
copy running-config tftp:
Address or name of remote host? 192.168.1.50
Destination filename? august_2020
```

What the command does: it copies the active running configuration out of the Cisco device and stores it on the TFTP server.

3. Restoring a Configuration from TFTP

A saved configuration can be pulled from the TFTP server back to the router or switch.

```
copy tftp: running-config
```

The device then asks for:

1. TFTP Server IP Address - the server that contains the backup.
2. Filename - the exact backup file that should be restored.

```
copy tftp: running-config
Address or name of remote host? 192.168.1.50
Source filename? august_2020
```

Key distinction: notice the direction of the copy operation. `running-config -> tftp:` is a backup, while `tftp: -> running-config` is a restore.

4. Backing Up the IOS Image

The IOS is stored in the device's flash memory. Before copying the image, the video first checks flash to identify the exact IOS filename.

```
show flash
```

Why this command comes first:

The copy command needs the exact source filename that exists in flash. Therefore, the filename shown by `show flash` is used in the next step.

After identifying the filename, the video starts the IOS backup with:

```
copy flash: tftp:
```

The device then requests:

1. Source filename - the exact IOS filename obtained from `show flash`.
2. TFTP server IP address - where the IOS backup should be stored.
3. Destination filename - the filename to use on the TFTP server.

Command flow

```
show flash  
copy flash: tftp:
```

5. Command Breakdown

Command	What it does
<code>copy running-config tftp:</code>	Backs up the active running configuration to a TFTP server.
<code>copy tftp: running-config</code>	Restores a saved configuration from TFTP into the running configuration.
<code>show flash</code>	Displays the contents of flash so the exact IOS image filename can be identified.
<code>copy flash: tftp:</code>	Copies the IOS image from flash memory to a TFTP server.

6. The Three Copy Patterns You Should Memorize

1) Configuration backup
`copy running-config tftp:`

2) Configuration restore
`copy tftp: running-config`

3) IOS image backup
`show flash`
`copy flash: tftp:`

Memory trick: read the command from left to right. The source is first, and the destination is second.

7. Practical Workflow

For a configuration backup:

```
running-config
|
v
TFTP server
```

For a configuration restore:

```
TFTP server
|
v
running-config
```

For an IOS backup:

```
flash
|
v
TFTP server
```

8. Important Details from the Video

- The TFTP server IP address is required during the copy operation.
- A destination filename is required when saving a configuration or IOS image to TFTP.
- When backing up IOS, first use `show flash` so you do not guess the source filename.

- When restoring a configuration, the source filename on the TFTP server must be the saved backup you intend to use.
- These operations are demonstrated for Cisco routers and switches.

9. Exam Notes

Remember the direction:

`copy running-config tftp: = device to TFTP`

`copy tftp: running-config = TFTP to device`

`copy flash: tftp: = IOS from flash to TFTP`

`show flash = find the IOS filename before copying it`

Do not skip: for an IOS backup, the video explicitly checks the flash filename first.

10. Quick Revision

```
BACKUP CONFIG
copy running-config tftp:
```

```
RESTORE CONFIG
copy tftp: running-config
```

```
CHECK IOS FILE
show flash
```

```
BACKUP IOS
copy flash: tftp:
```

The video content provided does not cover the full TFTP server setup, file permissions, IOS restore process, or boot-variable configuration. Those topics are therefore not added to this chapter.